

GENDER DIFFERENCES IN RISK BEHAVIOUR: DOES NURTURE MATTER?*

Alison L. Booth and Patrick Nolen

Using a controlled experiment, we investigate if individuals' risk preferences are affected by (i) the gender composition of the group to which they are randomly assigned, and (ii) the gender mix of the school they attend. Our subjects, from eight publicly funded single-sex and coeducational schools, were asked to choose between a real-stakes lottery and a sure bet. We found that girls in an all-girls group or attending a single-sex school were more likely than their coed counterparts to choose a real-stakes gamble. This suggests that observed gender differences in behaviour under uncertainty found in previous studies might reflect social learning rather than inherent gender traits.

It is well known that women are under-represented in high-paying jobs and in high-level occupations. Recent work in experimental economics has examined to what degree this under-representation may be due to innate differences between men and women. For example, gender differences in risk aversion, feedback preferences or fondness for competition may help to explain some of the observed gender disparities. If the majority of remuneration in high-paying jobs is tied to bonuses based on a company's performance, then, if men are less risk averse than women, women may choose not to take high-paying jobs because of the uncertainty. Differences in risk attitudes may even affect individual choices about seeking performance feedback or entering a competitive environment.

Understanding the extent to which risk attitudes are innate or shaped by environment is important for policy. If risk attitudes are innate, under-representation of women in certain areas may be solved only by changing the way in which remuneration is rewarded. However, if risk attitudes are primarily shaped by the environment, changing the educational or training context could help to address under-representation. Thus, the policy prescription for dealing with under-representation of women in high-paying jobs will depend upon whether the reason for the absence is innate to an individual's gender.

Why women and men might have different preferences or risk attitudes has been discussed but not tested by economists. Broadly speaking, those differences may be due to either nurture, nature or some combination of the two. For example, boys are pushed to take risks when participating in competitive sports, whereas girls are often encouraged to remain cautious. Thus, the riskier choices made by men could be due to

* Corresponding author: Alison L. Booth, Research School of Economics, College of Business & Economics, HW Arndt Bldg 25a, Australian National University, Canberra, ACT 0200, Australia. Email: al booth@essex.ac.uk

For helpful suggestions, we thank the editor, two anonymous referees, various seminar participants, and Uwe Sunde and Nora Szech. We also thank the students who participated in the experiment and their teachers who facilitated this. Financial support was received from the Australian Research Council, the British Academy, the Department of Economics at the University of Essex and the Nuffield Foundation. We take very seriously the ethical issues surrounding this research. This experiment received approval from the Ethics Committee of the University of Essex.

the nurturing received from parents or peers. Likewise, the disinclination of women to take risks could be the result of parental or peer pressure not to do so.

With the exception of Gneezy *et al.* (2009) and Gneezy and Rustichini (2004), the experimental literature on competitive behaviour has been conducted with college-age men and women attending coeducational universities. Yet, the education literature shows that the academic achievement of girls and boys responds differentially to coeducation, with boys typically performing better and girls worse than in single-sex environments (Kessler *et al.*, 1985; Brutsaert, 1999). Moreover, psychologists argue that the gendered aspect of individuals' behaviour is brought into play by the gender of others with whom they interact (Maccoby, 1998, and references therein). In this article, we sample a different subject pool to that normally used in the literature to investigate the role that environmental factors, which we label 'nurturing', might play in shaping risk preferences.¹ We use students in the UK from years 10 and 11 who are attending either single-sex or coeducational schools. We will examine the effect on risk attitudes of two types of environmental influences – randomly assigned experimental peer groups and educational environment. Finally, we will compare the results of our experiment with survey information – stated attitudes to risk obtained from a post-experiment questionnaire – to examine if reported and observed levels of risk aversion differ.

An important paper by Gneezy *et al.* (2009) explores the role that culture plays in determining gender differences in competitive behaviour. Gneezy *et al.* (2009) investigate two distinct societies – the patriarchal Maasai tribe of Tanzania and the matrilineal Khasi tribe in India. Although they find that, in the patriarchal society, women are less competitive than men – which is consistent with experimental data from Western cultures – in the matrilineal society, women are more competitive than men. Indeed, the Khasi women were found to be as competitive as Maasai men.² The authors interpret this as evidence that culture has an influence.³ We also use a controlled experiment to see if there are gender differences in the behaviour of subjects from two distinct environments or 'cultures'. But our environments – publicly funded single-sex and coeducational schools – are closer to one another than those in Gneezy *et al.* (2009) and it seems unlikely that there is much evolutionary distance between subjects from our two separate environments.⁴ Any observed gender differences in behaviour across these two distinct milieu is unlikely to be due to nature but more likely to the nurturing received from parents, teachers, peers or to some combination of these three.

Women are observed to be on average more risk averse than men, according to the studies summarised in Eckel and Grossman (2002).⁵ This could be through inherited

¹ In a companion paper, Booth and Nolen (2009; forthcoming), we investigate how competitive behaviour (including the choice between piece rates and tournaments) is affected by single-sex schooling.

² The experimental task was to toss a tennis ball into a bucket that was placed 3 metres away. A successful shot meant that the tennis ball entered the bucket and stayed there.

³ Interestingly, the authors find no evidence that, on average, there are gender differences in risk attitudes within either society.

⁴ We use subjects from two adjacent counties in south-east England, Essex and Suffolk. One would be hard pressed to argue that Essex girls and boys evolved differently from Suffolk girls and boys, popular jokes about 'Essex man' notwithstanding.

⁵ However, some experimental studies find the reverse. For example, Schubert *et al.* (1999), using as subjects undergraduates from the University of Zurich, show that the context makes a difference. Although women do not generally make less risky financial choices than men, they are less likely to engage in an abstract gamble.

attributes or nurture. The available empirical evidence suggests that parental attributes shape these risk attitudes. For instance, Dohmen *et al.* (2006; forthcoming) find, using the German Socioeconomic Panel (GSOEP), that individuals with highly educated parents are significantly more likely to choose risky outcomes. In this article, we will examine if, in addition to characteristics such as parental background, environment *also* plays a role in shaping risk preferences.

To examine if environment affects individuals' risk preferences, we study the choices made by students when they are randomly assigned to two different environments: same-sex or mixed-gender peer groups. Group-effects have been explored in previous work by Gneezy *et al.* (2003), Niederle and Yestrumskas (2008) and Datta Gupta *et al.* (2005) but those studies all used students from coeducational environments, focused on competitive tasks, and did not investigate risk attitudes. Besides looking at the effect of a peer group randomly created in the laboratory, we also examine how students from different educational environments – single-sex and coeducational schools – may have different risk preferences.

One of the strengths of our experiment is that we are able to look at an environment created in the laboratory – the experimental peer group – and one that is from the field – educational background. Although the experimental peer group is randomly assigned, students were not randomly assigned to school types. Therefore, to examine the effect of nurture from our second environment, we will have to deal with issues of selection. Our experiment was carefully constructed to deal with these. First, we sampled from two different counties – one with grammar schools and another without.⁶ Second, we designed the experiment so we could obtain good measures of cognitive ability in the early stages of the experiment. Third, we developed a post-experiment questionnaire to gather information on where students lived and their family background. The data from the questionnaire facilitate construction of a plausible instrument for single-sex school attendance. Fourth, we asked our participating coeducational schools to provide students only from the higher ability academic stream so that they would be more comparable with the grammar school students.⁷

Our final goal is to use the controlled experiment to see if commonly asked survey questions about risk yield the same conclusions about gender and risk aversion as those based on an experiment. During the experiment, our subjects can choose to make risky choices with real money at stake. At the end of the experiment, they answer questions about their risk attitudes as well as respond to a hypothetical lottery question.⁸ We are therefore able to compare actual behaviour with stated attitudes. This allows us to investigate (i) if girls and boys *behave* differently when there is actual money at stake; (ii) if girls and boys differ significantly in their *stated* attitudes to risk;⁹ (iii) if there are significant gender differences in the distance between actual choices made under

⁶ Grammar schools in the UK are selective state-funded schools for secondary school students.

⁷ To compare students of roughly the same ability, we recruited students from the top part of the distribution in the two coeducational schools in Essex: only students in the top academic streams from those schools were asked to participate. Students from Suffolk do not have the option to take the 11+ exam and therefore, higher ability students are unlikely to be selected out of Suffolk schools in the same way as in Essex. Nonetheless, we only recruited students from the top academic streams in Suffolk as well.

⁸ They are also asked questions about their family background, to be discussed later. These will form controls in the regression analyses.

⁹ For instance, boys might state that they are more risk-loving as a form of bragging.

uncertainty and stated behaviour under uncertainty; and (iv) if the general risk question is sufficient to describe actual risk-taking behaviour. In so doing, we explore the degree to which observed gender differences in choices under uncertainty and stated risk attitudes vary across subjects who have been exposed to single-sex or coeducational schooling. Furthermore, we are able to provide a comparison of results from a controlled experiment to commonly-asked survey questions.

1. Hypotheses

Women and men may differ in their propensity to choose a risky outcome because of innate preferences or because the existence of gender-stereotypes – well documented by Akerlof and Kranton (2000) – encourages girls and boys to modify their innate preferences. Our prior is that single-sex environments are likely to modify students' risk-taking preferences in ways that are economically important.

To test this, we designed a controlled experiment in which subjects were given an opportunity to choose a risky outcome – a real-stakes gamble with a higher expected monetary value than the alternative outcome with a certain payoff – and in which the sensitivity of observed risk choices to environmental factors could be explored. Suppose, there are preference differences between men and women. Then, using the data generated by our experiment to estimate the probability of choosing the real-stakes gamble, we should find that the female dummy variable is statistically significant. Furthermore, if any gender difference is due *primarily* to nature, the inclusion of variables that proxy the students' 'socialisation' should not greatly affect the size or significance of the estimated coefficient to the female dummy variable. However, if proxies for 'socialisation' are found to be statistically significant, this would provide some evidence that nurturing plays a role.

Our hypotheses can be summarised as follows.

CONJECTURE 1. *Women are more risk averse than men.*

As summarised in Eckel and Grossman (2002), most experimental studies have found that women are more risk averse than men. A sizable number of the studies used elementary and high-school-aged children from coed schools (Harbaugh *et al.*, 2002). Since our subject pool varies from this standard young adult sample, in that it involves students from both single-sex and coed schools, we will first examine whether or not there are gender differences in risk aversion. We expect to find that women in our sample are, on average, more risk averse than men.

CONJECTURE 2. *Girls in same-gender experimental groups are less risk averse than girls in mixed-gender experimental groups.*

Psychologists have shown that the framing of tasks and cultural stereotypes does affect the performance of individuals (see *inter alia* Steele *et al.*, 2002). Thus, a girl assigned to a mixed-gender experimental group may feel that her gender identity is threatened when she is confronted with boys. This might lead her to affirm her femininity by conforming to perceived male expectations of girls' behaviour, and consequently making less risky choices if she perceives risk avoidance as a feminine trait.

Should the same girl be assigned instead to an all-girl group, such reactions would not be triggered.

To test Conjecture 2, we randomly assign students to same or mixed-gender groups in the experiment. This allows us to examine if the gender composition of a group affects the risk preferences of girls. Since subjects are randomly assigned to groups, unobservables should not be driving the effects.

CONJECTURE 3. *Girls from single-sex schools are less risk averse than girls from coed schools.*

Studies show that there may be more pressure for girls to maintain their gender identity in schools where boys are present than for boys when girls are present (Maccoby, 1990; Brutsaert, 1999). In a coeducational environment, girls are more explicitly confronted with adolescent sub-culture (such as personal attractiveness to members of the opposite sex) than they are in a single-sex environment (Coleman, 1961). This may lead them to conform to boys' expectations of how girls should behave to avoid social rejection (American Association of University Women, 1992). If risk avoidance is viewed as being a part of female gender identity while risk-seeking is a part of male gender identity, then being in a coeducational school environment might lead girls to make safer choices than boys.

How might this actually work? It is helpful to extend the identity approach of Akerlof and Kranton (2000) to this context. Adolescent girls in a coed environment could be subject to more conflict in their gender identity, since they have to compete with boys academically while at the same time they may feel pressured to develop their femininity to be attractive to boys. Moreover, there may be an externality at work, since girls are competing with other girls to be popular with boys. This externality may reinforce their need to adhere to their female gender identity.

If this is true, we would expect girls in coed schools to be less likely than girls in single-sex schools to take risks. One might also expect coed schoolboys to be more likely to take risks than single-sex schoolboys, although the education literature suggests that there is greater pressure for girls to maintain their gender identity in schools where boys are present than for boys when girls are present (Maccoby, 1990, 1998).¹⁰

Given that subjects are not randomly assigned to a school, we will control for factors that could potentially be correlated with attendance, as will be explained later. Suppose we find that, conditional on observable factors, girls from single-sex schools choose to enter the tournament more than girls from coed school. This would provide more support for the case that nurture plays a role in determining risk preferences than if the controls explained all the difference in the choice whether or not to take a real-stakes gamble.

CONJECTURE 4. *Girls in same-gender environments (all-girl experimental groups or single-sex schools) are no less risk averse than boys.*

The psychological and education literature cited before suggests that girls, rather than boys, are likely to respond to the same-gender environments. The question is: how much will girls change? Given that we hypothesise that girls will be less risk averse

¹⁰ There is also evidence that in coed classrooms, boys get more attention and dominate activities (Sadker *et al.*, 1991; Brutsaert, 1999).

because of same-gender environments, we conjecture that girls' risk attitudes in single-sex environments will be the same as their male counterparts. If this is the case, it would suggest that the gender differences in observed risk attitudes are due to the environment and are not innate.

CONJECTURE 5. *Gender differences in risk aversion are sensitive to the way the preferences are elicited.*

To test this, we compare the results from the choice of whether or not to engage in a real-stakes gamble with responses obtained from two post-experiment survey questions. The first survey question is on general risk attitudes, whereas the second asks how much the respondent would invest in a risky asset using hypothetical lottery winnings.¹¹ Moreover, abstract real-stakes gambles might generate different gender gaps in risky choices than context-specific hypothetical gambles (Schubert *et al.*, 1999).

In particular, we examine how much, if at all, the answers about general risk attitudes or the hypothetical lottery explain observed choices made in the real-stakes experimental gamble. This allows us to examine how close stated risk attitudes are to observed behaviour and to see if girls and boys differ on any 'gap' that may exist. For example, suppose that boys state that they are more risk-seeking than they are in actuality, perhaps because being risk-loving is associated with a notion of 'hegemonic masculinity' governing male gender identity (Kessler *et al.*, 1985). If so, then boys might overstate their willingness to take risks when responding to a gender-attitudes survey question – after all, no real outcome depends on it – but be more likely to express their true risk aversion when confronted with a real-stakes gamble. In contrast, if being risk-loving is not part of female identity, there should be less distance in outcomes for girls.

2. Experimental Design

Our experiment was designed to test the Conjectures listed above. To examine the role of nurturing, we recruited students from coeducational and single-sex schools to be subjects. We also designed an 'exit' survey to elicit information about family background characteristics. At no stage were the schools we selected, or the subjects who volunteered, told why they were chosen. Our subject pool is relatively large for a controlled, laboratory-type experiment. We wished to have a large number of subjects from a variety of educational backgrounds to be able to investigate the Conjectures outlined before.

Next, we first discuss the educational environment from which our subjects were drawn, and then the experiment itself.

2.1. Subjects and Educational Environment

In September 2007, students from eight publicly funded schools in the counties of Essex and Suffolk in the UK were bussed to the Colchester campus of the University of

¹¹ Both questions will be given in full in the next Section.

Essex to participate in the experiment. Four of the schools were single-sex.¹² The students were from years 10 or 11, and their average age was just under 15 years. On arrival, students from each school were randomly assigned into 65 groups of four. Groups were of three types: all-girls, all-boys or mixed. Mixed groups had at least one student of each gender and the modal group comprised two boys and two girls. The composition of each group – the appropriate mix of single-sex schools, coeducational schools and gender – was determined beforehand. Thus, only the assignment of the 260 girls and boys from a particular school to a group was random. The school mix was two coeducational schools from Suffolk (103 students), two coeducational schools from Essex (45 students), two all-girl schools from Essex (66 students) and two all-boy schools from Essex (46 students).

In the county of Suffolk, there are no single-sex publicly funded schools. In the county of Essex, the old ‘grammar’ schools remain, owing to a quirk of political history.¹³ These grammar schools are single-sex and, like the coeducational schools, are publicly funded. It is highly unlikely that students themselves actively choose to go to the single-sex schools. Instead Essex primary-school teachers, with parental consent, choose the more able Essex children to sit for the Essex-wide exam for entry into grammar schools.¹⁴ Parents must be resident in Essex for their children to be eligible to sit the entrance examination (the 11+). However, residential mobility across regions is very low in Britain (Boheim and Taylor, 2002). To attend a grammar school, a student must apply and then attain above a certain score, which varies from year to year. Therefore, students at the single-sex schools are not a random subset of the students in Essex, since they are selected based on measurable ability at age 11.

One of the strengths of our experiment is that, although it does not solve the selection problem into single-sex and coeducational schools, it was carefully constructed to mitigate selection issues. First, we designed the experiment so we could obtain good measures of cognitive ability in the early stages of the experiment. These we then use as controls in the main part of the experiment. Second, we developed a post-experiment questionnaire, to gather information on where students lived and their family background. This facilitates construction of plausible instruments for school choice. Third, we asked our participating coeducational schools, from both Essex and Suffolk, to

¹² A pilot was conducted several months earlier, in June at the end of the previous school year. The point of the pilot was to determine the appropriate level of difficulty and duration of the actual experiment. The pilot used a different subject pool to that used in the real experiment. It comprised students from two schools (one single-sex in Essex and one coeducational in Suffolk) who had recently completed year 11. The actual experiment conducted some months later, at the start of the new school year, used, as subjects, students who had just started years 10 or 11.

¹³ In the UK, schools are controlled by local area authorities but frequently ‘directed’ by central government. Following the 1944 Education Act, grammar schools became part of the central government’s tripartite system of grammar, secondary modern and technical schools (the latter never got off the ground). By the mid-1960s, the central Labour government put pressure on local authorities to establish ‘comprehensive’ schools in their place. Across England and Wales, grammar schools survived in some areas (typically those with long-standing Conservative boroughs) but were abolished in most others. In some counties, the grammar schools left the state system altogether and became independent schools; these are not part of our study. However, in parts of Essex, single-sex grammar schools survive as publicly funded entities, whereas in Suffolk, they no longer exist.

¹⁴ If a student achieves a high enough score on the exam, s/he can attend one of the 12 schools in the Consortium of Selective Schools in Essex (CSSE). The vast majority of these are single-sex. The four single-sex schools in our experiment are part of the CSSE.

provide students only from the higher ability academic stream so that they would be more comparable to the grammar school students, as noted in footnote 7.

The experiment took place in a very large and spacious auditorium, with 1,000 seats arranged in tiers.¹⁵ Students in the same group were seated in the same row with an empty seat between each person. There was also an empty row in front of and behind each group. Although subjects were told which other students were in the same group, they were sitting far enough apart for their work to be private information. If two students from the same school were assigned to a group, they were forced to sit as far apart as possible; for example, in a group of four, two other students would sit between the students from the same school. There was one supervisor, a graduate student, assigned to supervise every five groups. Once the experiment began, students were told not to talk. Each supervisor enforced this rule and also answered individual questions.

2.2. *Experiment*

Five rounds were conducted during the experiment. In Appendix A, we give full details of all rounds and there we also describe payments and incentives, which varied from round to round.¹⁶ In this article, we focus on the results from the round involving the real-stakes gamble or *fiver lottery*. After the experiment ended, students filled out a post-experimental questionnaire that had questions on risk attitudes, family background, and that also included a hypothetical investment decision using the proceeds from winning a lottery.¹⁷

A description of the real-stakes gamble (called the ‘fiver’ lottery) and the two main survey questions are discussed below.

‘Fiver’ Lottery. Each student chooses Option 1 or Option 2. Option 1 is to get £5 for certain. Option 2 is to flip a coin and get £11 if the coin comes up heads or get £2 if the coin comes up tails.

Survey Question: General Risk. Each student was asked: ‘How do you see yourself: Are you generally a person who is fully prepared to take risks or do you try to avoid taking risks?’ The students then ranked themselves on an 11-point scale from 0 to 10 with 0 being labelled ‘risk averse’ and 10 as ‘fully prepared to take risks’. This was the same general risk question asked in the 2004 wave of the GSOEP.

Survey Question: Hypothetical Lottery. Each student was asked to consider what they would do in the following situation: ‘Imagine that you have won £100,000 in the lottery. Almost immediately after you collect the winnings, you receive the following financial offer from a reputable bank, the conditions of which are as follows: (i) there is a the chance to double your the money within two years; (ii)

¹⁵ Students were very clear about who was in their group and – while nothing was explicitly stated – they could see if their group was all-girls, all-boys or mixed. Other studies, such as Gneezy and Rustichini (2004), also look at the effect of single-sex environments by making the reference group obvious while not quarantining the students.

¹⁶ Payment was randomised in the same manner as in Datta Gupta *et al.* (2005) and Niederle and Vestergaard (2007). As students are paid for a round that is randomly selected at the end of the experiment, each individual should maximise her/his payoff in each round to maximise the payment overall. Moreover, as only one round was selected for payment, students did not have the opportunity to hedge across tasks.

¹⁷ Results from the first few rounds, designed to elicit differences in competitive behaviour under piece rates and tournaments, are reported in Booth and Nolen (2009; forthcoming).

it is equally possible that you could lose half the amount invested; (iii) you have the opportunity to invest the full amount, part of the amount or reject the offer. What share of your lottery winnings would you be prepared to invest in this financially risky yet lucrative investment?' The subject then ticked a box indicating if she would invest £100,000, £80,000, £60,000, £40,000, £20,000 or Nothing (reject the offer). The same version of this hypothetical investment question was asked in the 2004 wave of the GSOEP.

The payments (both the show-up fee of £5 plus any payment from performance in the randomly selected round) were in cash and were hand-delivered in sealed envelopes (clearly labelled with each student's name) to the schools a few days after the experiment. The average payment was £7. In addition, immediately after completing the Exit Questionnaire, each student was given a bag containing a soft drink, packet of crisps and bar of chocolate.

2.3. Descriptive Statistics

We will be examining risk preferences by experimental peer group and schooling type. Since the experimental peer group was randomly assigned, we expect that observables should not differ by group type: same or mixed gender. However, since school type was not randomly assigned, we will control for ability, learning and any other background attribute that could be drive the schooling result. Table 1 shows girls' and boys' summary statistics by experimental peer group and school type.

Panels (a) and (c) of Table 1 compare the means of same-gender experimental groups (all-girls or all-boys) with the mixed-gender groups, for girls and boys, respectively. There are no statistical differences – except for age for girls and the number of siblings for boys – suggesting that the randomisation was implemented successfully. Therefore, when examining risk preferences in the following Section, we control for age and number of siblings in some specifications. Note also that there is no statistically significant difference in means for risk choice (the Fiver Lottery) for either girls or boys. However, panel (a) shows that the difference for girls is 0.12 and this is significant at the 11% level.

Now consider panels (b) and (d) of Table 1, which compare the means of students at single-sex and coed schools for girls and boys, respectively. Inspection reveals a number of observable differences. For instance, both girls and boys at single-sex schools are more likely to have parents who went to university. Girls at single-sex schools are less risk averse than their coed counterparts (they are more likely to choose the 'fiver' lottery and to report a greater willingness to take risks). It is interesting that this is not the case for boys. Boys at single-sex schools are likely to be older than boys at coed schools and girls at single-sex schools have fewer siblings than girls at coed schools. When examining the effect of schooling environment, we will control for these observed differences in some specifications.

3. Experimental Results

In this Section, we discuss whether or not the results from the fiver lottery support the first four Conjectures. We then use a series of robustness checks to see, first, if the

Table 1
Descriptive Statistics by Gender, Experimental Peer Group and School Background

Variables	Girls				Boys			
	(a) Experimental group		(b) Schooling		(c) Experimental group		(d) Schooling	
	Mixed	All-girls	Difference	Coeducation	Single-sex	Difference	Coeducation	Single-sex
Chose 'Five' lottery	0.61	0.73	0.12	2.16	2.62	0.46***	2.88	3.13
Round 1 score	2.38	2.32	-0.06	0.54	0.86	0.32***	0.88	0.78
Round 2 score	3.92	3.93	0.01	3.78	4.14	0.36	4.71	5.17
Number of siblings	0.84	1.04	0.2	0.88	1.05	0.17	0.85	0.61
Number of female siblings	0.70	0.71	0.01	0.80	0.57	-0.23*	0.87	0.68
Age	14.73	14.98	0.25***	14.80	14.95	0.15	14.81	14.48
Mother went to uni (=1)	0.28	0.26	-0.02	0.13	0.49	0.36***	0.15	0.43
Father went to uni (=1)	0.28	0.31	0.03	0.16	0.52	0.36***	0.27	0.54
Coed school travel time (min)	7.92	6.89	-1.03	6.21	8.06	1.85*	7.16	6.8
Single-sex school travel time (min)	18.80	17.47	-1.33	23.09	15.32	-7.77***	21.95	12.96
Average risk score (Scale = 0-10)	6.73	6.54	-0.19	6.40	6.95	0.55*	6.90	6.69
Hypothetical lottery investment (£1,000)	3.24	3.47	0.23	2.76	4.24	1.48***	3.73	3.48

Notes. Significant differences are denoted: *significant at a 10% level; **significant at a 5% level; ***significant at a 1% level.

evidence stands up to the use of different control groups and, second, if the results alter when we instrument for single-sex schooling.

3.1. 'Fiver' Lottery

The expected monetary value of the fiver lottery discussed before is £6.50, which is greater than the alternative choice – a certain outcome of £5. Assuming a constant relative risk aversion utility function of the type $u(x) = x^{1-\sigma}/(1-\sigma)$, where σ is the degree of relative risk aversion, we calculate that the value of σ making an individual just indifferent between choosing the lottery and the certain outcome is approximately 0.8. Individuals with $\sigma \geq 0.8$ will choose the certain outcome, whereas those with $\sigma < 0.8$ will choose the lottery.¹⁸

To examine if there are any gender differences in the choice of whether or not to enter the fiver lottery, we construct an indicator variable that takes the value one if the individual chooses to enter the fiver lottery and zero otherwise. This becomes our dependent variable in a probit model of the probability of choosing the lottery. Table 2 shows the marginal effects of those probit regressions.

The first column of Table 2 shows that, on average, girls choose to enter the lottery 16 percentage points less than boys. The sign and significance of this coefficient is consistent with other work looking at gender and risk aversion and suggests that, in our sample, female students are also more risk averse than male students. This provides evidence for Conjecture 1. Next, we want to investigate if the gender differences alter when environmental factors reflecting nurture are incorporated into the estimation.

The specification in column [2] adds controls for school type and experimental group composition. In this specification, the gender gap becomes even more pronounced – girls in coed schools choose to enter the lottery 36 percentage points less than boys from coeducational schools. Furthermore, we have evidence that nurture has an effect on risk preferences. First, the coefficient for being in an all-girls group is statistically significant and positive: girls randomly assigned to all-girl groups are more likely to choose to enter the 'fiver' lottery. Because our estimates show that girls assigned to single-sex peer groups are less risk averse than those who are assigned to mixed-gender groups, evidence is provided in support of Conjecture 2.¹⁹ Of note is that the same-gender peer group is only affecting girls; the all-boys coefficient is insignificant. Second, the single-sex school coefficient is statistically insignificant but the coefficient to single-sex schooling interacted with female is significant and positive. Therefore, school background only affects the risk preferences for girls at this level of risk aversion, and has no effect on boys. The risk preferences of boys are not affected by either environmental variable, whereas the risk preferences of girls are significantly

¹⁸ This was calculated from $pu(x) + (1-p)u(y) = u(z)$, where $p = 0.5$, $x = 11$, $y = 2$ and $z = 5$. We use the specific CRRA functional form for $u(\cdot)$ given in the text. Our imposed value of σ is higher than that at which German adults switch from a safe to an uncertain outcome found by Dohmen *et al.* (2010). See also Holt and Laury (2002).

¹⁹ The all-girls coefficient is robust to different types of analysis. For instance, if regressions are run on subsamples comprising only students from coed schools, or only students from single-sex schools, the all-girl coefficient is still significant. Thus, there is a positive effect of being in an all-girls group for each type of student. Furthermore, if dummy variables for mixed-gender groups with three or two boys are used as controls, the significance of the all-girls coefficient does not go away.

Table 2
Dependent Variable (=1) If Student Chose Option 2 in 'Fiver' Lottery

Coefficient	[1]	[2]	[3]	[4]	[5]	[6]	[7]
Female (=1)	−0.16*** (0.05)	−0.36*** (0.07)	−0.37*** (0.07)	−0.34*** (0.11)	−0.39*** (0.09)	−0.43*** (0.08)	−0.46*** (0.09)
Single-sex (=1)		−0.13 (0.10)	−0.13 (0.10)	−0.06 (0.18)	−0.21** (0.10)	−0.10 (0.08)	−0.11 (0.10)
Female × single-sex		0.33*** (0.06)	0.33*** (0.06)	0.30** (0.12)	0.38*** (0.07)	0.42*** (0.10)	0.51*** (0.14)
All-girls group (=1)		0.12* (0.06)	0.12* (0.06)	0.14** (0.06)	0.11* (0.06)	0.13* (0.07)	0.13* (0.07)
All-boys group (=1)		−0.05 (0.10)	−0.04 (0.10)	−0.05 (0.11)	−0.00 (0.10)	−0.04 (0.08)	−0.04 (0.08)
Maze score R1			−0.01 (0.03)				
Maze score R2 − R1			0.02 (0.02)				
Marginal effect for female = Single-sex = female × single-sex = 1		−0.07 (0.05)	−0.06 (0.05)	−0.03 (0.06)	−0.10 (0.07)		
Controls	No	No	No	Yes	No	No	No
Controls × female	No	No	No	Yes	No	No	No
Controls × single-sex	No	No	No	Yes	No	No	No
Controls × female × single-sex	No	No	No	Yes	No	No	No
Model	Probit	Probit	Probit	Probit	Probit	LPM	IV LPM
Constant						0.90*** (0.05)	0.90*** (0.06)
Observations	260	260	260	260	204	260	260
R ²						0.131	0.107
F-stat for IV variables							118.9

Notes. Columns [1]–[4] and [6]–[7] use entire sample. Column [5] uses only students from single-sex schools, students who took 11+ examination, and students from Suffolk. Controls: mother went to University (=1); father went to University (=1); number brothers; sisters; student aged 14 (=1). Robust SEs in parenthesis; ***p < 0.01, **p < 0.05, *p < 0.1.

affected by both environmental factors. This provides strong evidence for conjecture three.

We now check the robustness of the results in support of Conjectures 2 and 3. We do this by looking at the effect of cognitive skills, by using different sub-samples and by saturating our regression with all observable differences that were found in Table 1.

3.1.1. Sensitivity analysis

Why should cognitive skills affect individuals' economic preferences? ²⁰ Differences in perception of risky options due to cognitive ability may systematically affect individuals' choices, as argued by Burks *et al.* (2009). The more complex is an option, the larger the noise. If people of high cognitive skills perceive a complex option more precisely than people with low cognitive skills, they will be more likely to choose riskier options. How do our estimates alter when we add in these measures of cognitive ability? When controlling for ability and performance in the previous rounds, as in column [3] of

²⁰ Dohmen *et al.* (2010), using a random sample of around 1,000 German adults, found that lower cognitive ability is associated with greater risk aversion and more pronounced impatience. A similar result was found in by Burks *et al.* (2009), with a sample of 1,000 trainee truckers in the US.

Table 2, the all-girls coefficient does not change or become insignificant. Likewise, there is little change to the coefficient for single-sex education or for single-sex education interacted with being female.²¹

To examine if observable differences between the samples are driving our results, we saturate our preferred specification. Column [4] of Table 2 includes all controls whose means differed across sub-groups in Table 1, and their interactions with female, single-sex, and the female, single-sex interaction. After controlling for observables and their interactions with schooling background, we find that the coefficients only change slightly. In fact, the randomly assigned variable, all-girls group, becomes more significant but the single-sex female interaction changes slightly. This, again, supports Conjecture 2. However, given the slight change in the female, single-sex interaction, we next investigate more closely the evidence in favour of Conjecture 3.

To test the robustness of the female, single-sex coefficient, we first estimate the model on a different sub-group and second, we use an instrumental variable. Initially, we discuss the results from estimation on different sub-groups.

Earlier we noted that a student's attendance at a single-sex school is likely to be influenced by her ability as well as by the choices of her parents or teachers.²² Therefore, students from single-sex schools may not be a random subset of the students from Essex. However, it should be remembered that we asked only top students from coeducational schools to participate in the experiment. As a sensitivity analysis, we compare single-sex students to a different comparison group: students from Suffolk *plus* students in Essex who took the 11+ exam. Column [5] of Table 2 is estimated on a sub-sample comprising students from Suffolk, students who took the 11+ examination, and students from single-sex schools. Students in Suffolk have to attend their closest school so they are likely to be a more representative sample. Furthermore, if 'parental pushiness' is an issue, then those students who took the 11+ examination should look more like the single-sex students. Using this sub-sample, we see that the gender gap actually becomes slightly larger: girls are 39 percentage points less likely to enter the lottery. However, the coefficient to single-sex schooling is also negative and significant. This suggests that boys in coed schools are more likely to take risks and perhaps 'show off' for the girls (that stereotype threat could be causing the gender gap in risk aversion to be larger). This evidence would fit with the discussion of Conjecture 2. Girls in all-girl groups are again more likely to enter the lottery than girls in coed groups, also providing evidence for Conjecture 2. Notice also that the interaction of single-sex schooling and female, although remaining significantly negative, becomes slightly larger in absolute terms.

Now, we turn to our final robustness check. To control for the fact that students at single-sex schools are a non-random subset of the student population, we instrument for single-sex school attendance using the difference in travel time between the closest

²¹ To see if the impact of performance in the first two rounds differed by gender, we also experimented with interacting maze score from R1 and maze score R2 – R1 with gender and this did not change our estimates of interest. As mentioned earlier, since the round that was paid was randomly chosen, we would not expect effort to change across rounds.

²² As noted earlier, Essex primary-school teachers and parents choose which children sit for the Essex-wide exam for entry into grammar schools. Parents must be resident in Essex for their children to be eligible to sit the entrance examination (the 11+).

coed and single-sex school. First, we present the regression results of the linear probability model (LPM) in column [6] of Table 2, whereas the results for the IV are in column [7] of Table 2. We used the six-digit residential postcode for each student to calculate the distances to the nearest single-sex school and to the nearest coed school.²³ We next calculated a variable equal to the minimum time needed to travel to the closest single-sex school minus the minimum time to travel to the closest coeducational school. We break this variable into two instruments: difference in minutes of travel time if difference in travel time is less than the average; and difference in minutes of travel time if difference in travel time is more than the average. We then instrumented for attendance at a single-sex school using a two-step process. First, we estimated the probability of a student attending a single-sex school, where the explanatory variables were an Essex dummy (taking the value one if the student resides in Essex and zero otherwise) and an interaction of Essex-resident with the two travelling-time variables. We then estimated the regression reported in column [7], which is a LPM, where we use predicted single-sex school attendance in place of the original single-sex school dummy.²⁴ Since the equation uses predicted values, we bootstrapped the standard errors for attending a single-sex school.²⁵ Even here we find that the female, single-sex schooling interaction and all-girls group variable are statistically significant; indeed, the coefficient to the interaction of female with single-sex schooling is even larger.

Given these robustness checks, we therefore conclude that there is strong evidence for Conjecture 3 – that the schooling environment can affect risk preferences.

Conjecture 4 was that girls in single-sex environments (all-girls groups or single-sex schools) would be just as likely to enter the lottery as boys. In each column [2–5] in Table 2, we presented the marginal effect for a girl in a single-sex school compared to a boy in a coed school. In each case, the estimated effect is insignificant. However, if one were to take the point estimates seriously, then the single-sex schooling environment has reduced the gender gap by over 70% in all cases. The all-girls group coefficient is not as large as the female, single-sex interaction and, therefore, there is still a significant gap when groups are controlled for. Thus, we have mixed evidence for Conjecture 4.

Given these robustness checks and the continued significance of the all-girls group variable and the single-sex, female interaction, there seems to be strong evidence for Conjectures 2 and 3, that girls in same-gender groups enter the lottery more than girls in mixed-gender groups and that single-sex girls enter the lottery more than coed girls. There is also evidence for part of Conjecture 4, that girls in single-sex environments take the risky option as much as boys. The marginal effects for single-sex girls compared with coed boys is negative in all columns of Table 2 but they are insignificant,

²³ To calculate this, we used the postcode of each school and the postcode in which a student resides. We then entered the student's postcode in the 'start' category in MapQuest.co.uk (<http://www.mapquest.co.uk/mq/directions/mapbydirection.do>) and the school's postcode in the 'ending address'. Mapquest then gave us a 'total estimated time' for driving from one location to the other. It is this value that we used. Thus, the 'average time' is based on the speed limit of roads and the road's classification (i.e. as a motorway or route).

²⁴ The first-stage results for the IV regression are included as Table A1. The F-statistic for the instruments is 118.9 and is listed at the bottom of column [7] in Table 2. Note that we do not enter travel time linearly because the data showed a non-linear trend.

²⁵ We randomly drew 1,000 different samples from our experimental data to calculate the bootstrap results.

suggesting that single-sex girls choose the risky option as much as boys. However, the size of the coefficient on the all-girls group dummy variable is not large enough to cancel out the negative coefficient on the female dummy variable. Therefore, girls in same-gender groups are not entering the lottery as much as coed boys. Since girls in some same-gender environments are not choosing the risky option as much as boys, then we cannot fully support this hypothesis. The length of time a girl has been exposed to the same-gender environment – three years on average for girls at single-sex schools and only 30 minutes for girls in single-sex groups – may explain the difference in the size of the effect. However, the support for Conjectures 2, 3 and part of 4 provides strong evidence that nurture is affecting the risk attitudes of girls and also that the magnitude of this effect is large; completely cancelling out the gender gap in some cases. We next examine if this finding can also be revealed using commonly asked survey questions about risk.

4. Survey *Versus* Experimental Results

The experimental setting provided evidence that nurturing affects a girl's behaviour under uncertainty. We now examine whether survey questions could have been used to obtain those results and if the answers to commonly used survey questions provide any predictive power in explaining how a subject behaves in an experimental setting. To see if a student's answer to the general risk question, outlined in detail in Section 3.2, provided any insight into whether the student would enter the fiver lottery, we reran that probit regression with an additional control for general risk attitude. The marginal effects are reported in column [2] of Table 3. The results show that choosing the fiver lottery is positively correlated with how prepared a student is to take risks. But inclusion of risk attitudes does not take away the explanatory power of the single-sex, female interaction or of the all-girls group coefficient.²⁶ Furthermore, the interaction of responses to the general risk question with being female is statistically insignificant. If student responses to the general risk attitudes question pick up their unobserved propensity to overstate their risk-loving, then the insignificance of this interaction implies that neither sex overstates more than the other.

Likewise, when we use the student's answer to the hypothetical lottery – column [3] of Table 3 – the explanatory power of the single-sex female interaction and being in all-girls group coefficient remain statistically significant even though some of the coefficients to the survey response are also statistically significant. However there is little difference in how boys and girls responded to the survey question, as the hypothetical amount interacted with being female has little explanatory value. This again suggests that the survey questions are being answered in a similar way by both boys and girls.

Since the general risk question and the answer to the hypothetical lottery are typically positively correlated with choosing to enter the fiver lottery, we now examine if the answers to the two survey questions could have been used as dependent variables

²⁶ This result is robust to entering a dummy variable for each option in the general risk scale, or for entering a squared term for the general risk question.

Table 3
Examining the Experimental and Survey Results

Variables	Dependent variable							
	(=1) If a student chose option two in 'Fiver' Round			Readiness to take Risk (0–10)		Hypothetical lottery outcome		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female (=1)	–0.36*** (0.07)	–0.51*** (0.13)	–0.33*** (0.08)	–0.23 (0.21)	–0.44 (0.37)	–0.79 (0.58)	–0.92 (0.75)	–0.29 (0.23)
Single-sex (=1)	–0.13 (0.10)	–0.12 (0.10)	–0.13 (0.10)	–0.09 (0.22)	–0.21 (0.40)	–0.34 (0.56)	–0.31 (0.71)	–0.11 (0.22)
Female × single-sex	0.33*** (0.06)	0.32*** (0.06)	0.28*** (0.06)	0.37 (0.27)	0.75 (0.50)	1.83** (0.71)	2.25** (0.89)	0.70*** (0.27)
All-girls (=1)	0.12* (0.06)	0.13** (0.06)	0.12** (0.05)	–0.09 (0.16)	–0.18 (0.29)	0.25 (0.43)	0.29 (0.55)	0.09 (0.16)
All-boys (=1)	–0.05 (0.10)	–0.06 (0.11)	–0.04 (0.10)	–0.07 (0.23)	–0.12 (0.42)	1.00 (0.61)	1.24* (0.75)	0.38* (0.23)
Readiness to take risk (0–10)		0.04* (0.02)						
Female × readiness to take risk		0.04 (0.03)						
Invest £20,000 (=1)			–0.07 (0.14)					
Invest £40,000 (=1)			0.06 (0.11)					
Invest £60,000 (=1)			0.04 (0.12)					
Invest £80,000 (=1)			0.34*** (0.04)					
Invest £100,000 (=1)			–0.14 (0.28)					
Female × invest £20,000 (=1)			0.05 (0.13)					
Female × invest £40,000 (=1)			–0.13 (0.18)					
Female × invest £60,000 (=1)			0.16** (0.08)					
Female × invest £80,000 (=1)			–0.88*** (0.02)					
Female × invest £100,000 (=1)			0.13 (0.12)					
Model type	Probit	Probit	Probit	Ordered probit	OLS	OLS	Tobit	Ordered probit
Constant					6.94*** (0.27)	3.41*** (0.45)	2.87*** (0.59)	
Observations	260	255	259	255	255	259	259	259
R ²					0.019	0.058		

Notes. Cutoffs for the ordered probit regression in column [4] are: –2.79, –2.55, –1.65, –1.27, –0.75, –0.46, 0.32, 0.86, and 1.52. Cutoffs for the ordered probit regression in column [8] are: –0.72, –0.05, 0.64, 1.22, 1.9. All cutoffs from both ordered probit regressions are significant at the 1% level except for –0.05 which is insignificant. Robust standard errors in parenthesis. ***p < 0.01, **p < 0.05, *p < 0.1.

instead of the real-stakes experimental outcome.²⁷ Column [4] of Table 3 uses the responses to the general risk question as the dependent variable. In this case, the regression model used is Ordered Probit. Notice that all of the variables of interest are now statistically insignificant. There is no gender effect (the female dummy is not significant); there is no school-level nurturing (the single-sex and female interaction is insignificant) and there is no effect of the experimental peer group. Even if OLS is used – as done in column [5] of Table 3 – or a binary variable is created from the general risk attitudes question – using any cut point ranging from 3 to 8 – the survey question does not yield the same results as the real-stakes experimental lottery.

Columns [6–8] of Table 3 use the responses to the hypothetical risky financial investment as the dependent variable. As noted earlier, this not only represents a risky investment decision, as distinct from the abstract gamble for real stakes represented by the fiver lottery, but it also involves hypothetical amounts. Column [6] reports the results from OLS estimation, allowing a straightforward interpretation of the results. Notice that the female dummy has a statistically insignificant effect but that the interaction of female with single-sex schooling is statistically significant at the 5% level. These estimates suggest that, *ceteris paribus*, girls from single-sex schools are willing to invest more than boys from coed schools. In other words, they invest nearly one and a half units more (where each unit involves an increase in the money the individual would invest in the lottery of around £20,000). They also invest significantly more than coed girls.

As a comparison, in column [7], we report results from a tobit model (used because a student can choose to put none of her hypothetical lottery winnings in the risky investment). Again, only the single-sex school and female interaction is significant at the 5% level. The magnitude is larger than in the column of OLS estimates. Finally, column [8] reports results from an ordered probit model and again only the single-sex and female result is significant (5% level). This suggests that, while the hypothetical lottery investment does not provide the same evidence about relative risk aversion as the real-stakes experimental lottery, nonetheless the interaction of female with single-sex schooling remains positive and statistically significant across the three estimation methods.

In summary, using the survey question as the dependent variable would suggest that, while there is no gender difference in risk aversion, women attending single-sex schools are not only as likely as men to enter the real-stakes gamble but they also invest more in the hypothetical risky investment than do coed women and all men.

Given the results in Table 3, it seems that there is mixed evidence for Conjecture 5. Although the commonly used general risk attitude question is positively correlated with actual risky choices made under uncertainty, the determinants of these general risk attitudes differ quite markedly from the determinants of *actual* risky choices under uncertainty. This suggests that relying only on general risk attitudes might lead

²⁷ *A priori*, we would expect different responses to the risk attitudes and the hypothetical lottery questions. The hypothetical lottery question is not as straightforward as the simple risk attitudes questions because it involves not only a time dimension (the outcome will not be realised until two years hence) and uncertain returns but also a choice of investment amounts. Risk aversion and impatience are thus conflated in responses to this question, as discussed in Booth and Katic (2011). The fact that it is more complex may also raise narrow bracketing issues, whereby ‘a decision maker who faces multiple decisions tends to choose an option ... without full regard to the other decisions and circumstances that she faces (Rabin and Weizsäcker, 2009, p. 1508).

researchers to make misleading inferences about gender differences in choice under uncertainty. In contrast, the determinants of the amounts invested from the hypothetical lottery had some similarities to the determinants of actual risky choices under uncertainty. Estimating the determinants of amounts invested from the hypothetical lottery yielded the insight that girls attending single-sex schools invest more in the risky outcome than boys. The real-stakes experimental lottery showed that girls from single-sex school were as likely as boys to enter the lottery, which is not inconsistent with the hypothetical lottery results. This example illustrates the complementary roles of experimental and survey data and suggests that gender differences in risk aversion differ across contexts.

5. Conclusion

Women and men may differ in their propensity to choose a risky outcome because of innate preferences or because pressure to conform to gender-stereotypes encourages girls and boys to modify their innate preferences. Single-sex environments are likely to modify students' risk-taking preferences in economically important ways. To test this, we designed a controlled experiment in which subjects were given an opportunity to choose a risky outcome – a real-stakes gamble with a higher expected monetary value than the alternative outcome with a certain payoff – and in which the sensitivity of observed risk choices to environmental factors could be explored. The results of our real-stakes gamble show that gender differences in preferences for risk-taking are indeed sensitive to whether the girl attends a single-sex or coed school. Girls from single-sex schools are as likely to choose the real-stakes gamble as boys from either coed or single-sex schools, and more likely than coed girls. That girls in co-educational institutions have a lower preference for risk is interesting, since this is the norm for children in schools in many countries including the UK.

Moreover, we found that gender differences in preferences for risk-taking are sensitive to the gender mix of the experimental group, even when these groups are placed within a larger coeducational milieu. In particular, we found that girls are more likely to choose risky outcomes when assigned to all-girl groups. This finding is relevant to the policy debate on whether or not single-sex classes within coed schools could be a useful way forward.

We also found that gender differences in risk aversion are sensitive to the method of eliciting preferences. While the commonly used general risk attitude question is positively correlated with actual risky choices made under uncertainty, the determinants of these general risk attitudes differ quite markedly from the determinants of actual risky choices under uncertainty. This suggests that to rely only on survey-based general risk attitudes might lead researchers to make misleading inferences about gender differences in choice under uncertainty. In contrast, the determinants of the amounts invested from the hypothetical lottery had some similarities to the determinants of actual real-stakes gambles under uncertainty.

To summarise our main results, on average girls from single-sex schools are found in our experiment to be as likely as boys to choose the real-stakes lottery. This suggests that observed gender differences in behaviour under uncertainty found in previous studies might reflect social learning rather than inherent gender traits, a finding that

would be hard, if not impossible, to show, using survey-based evidence alone. We hope that future work will further explore these issues in situations where there has been random assignment of students to single-sex education, a policy that some jurisdictions are currently contemplating introducing.

Finally, we note that our experiment does not allow us to tease out why these behavioural responses were observed for girls in all-female environments. In an interesting recent paper, Lavy and Schlosser (2011) found that an increase in the proportion of girls improves boys' and girls' cognitive outcomes in mixed-gender classes. They attributed this to lower levels of classroom disruption and violence, improved inter-student and student–teacher relationships, and reduced teachers' fatigue. However, this mechanism does not address the single-sex school effects that we have found in our analysis. Conjectures as to why girls in single-sex schools have different risk preferences to their coed counterparts, as we have found, might include the following. Adolescent females, even those endowed with an intrinsic propensity to make riskier choices, may be discouraged from doing so because they are inhibited by culturally driven norms and beliefs about the appropriate mode of female behaviour – avoiding risk. But once they are placed in an all-female environment, this inhibition is reduced. No longer reminded of their own gender identity and society's norms, they find it easier to make riskier choices than women who are placed in a coed class. We hope in future research to further investigate these hypotheses.

Appendix A. The Experiment

In the experiment, students were escorted into a large auditorium. One individual read aloud the instructions to everyone participating. All the graduate supervisors hired to supervise groups were given a copy of the instructions, were involved in the pilot that had taken place, and had gone through comprehensive training. These supervisors answered questions if they were raised.

Below is the text of the slides that were shown to the students when they arrived at the auditorium:

Slide 1:

Welcome to the University of Essex!

Today you are going to be taking part in an economics experiment.

Treat this as if it were an exam situation:

No talking to your neighbours.

Raise your hand if you have any questions.

There will be no deception in this experiment.

Slide 2:

The experiment today will involve completing three rounds of mazes.

Rules for completing a maze:

Get from the flag on the left-hand side to the flag on the right-hand side.

Do not cross any lines!

Do not go outside of the box.

We will now go through an example!!

Comment:

At this point students were shown one practice maze and were walked through how to solve it, illustrating the three points raised above.

Slide 3:

The supervisors in your row will be handing you maze packets throughout the session.

At all times you need to put your seat letter and number on the packet and your name.

Please make sure you know your row letter and seat number.

Your seat is also on your badge. It is the middle grouping. For example, if your badge was 1-A3-F your seat number should be A3. Make sure this is correct now.

Mazes: You should do the mazes in order.

If you cannot solve a maze put an X through it and go onto the next maze.

If you do not put an X through it none of the following mazes will be marked.

Note: If you do not have the correct seat number on your maze packets you may be paid incorrectly.

Slide 4:

We are going to be doing six rounds of mazes.

Before each round of mazes we will explain how you will be paid for that round.

After all six rounds of mazes are finished we will choose one round to 'implement'.

That means you will get paid for your performance in that round.

The round for which you will be paid will be chosen randomly from this cup.

You will also receive GBP 5 for showing up today.

Since you do not know for which round you will be getting paid, you should do your best in each round and treat each round separately.

Slide 5:

You will get 5 min to solve up to 15 mazes.

Please solve as many mazes as you can.

Do not begin until I say go!

For this round you will get £0.50 *for each maze you solve correctly*.

Example: If you solve eight mazes correctly you will earn GBP 4.

Please make sure you have put your name and seat on the maze packet now.

Are there any questions?

OK → GO!

OK → STOP

No Talking!

Slide 6:

Now you will get £2 *for each maze you solve correctly* IF *you solve the most mazes correctly in your group*.

Your group consists of you and the three other people sitting in your 'row' who have the same first number on their badge.

Example: If your badge number is 1-B2-M then your group consists of you and the three other students with the badges 1-*** in your row.

If you are in group 1 and you solve eight mazes correctly then:

IF everyone else in your group solved fewer than eight mazes correctly you will get GBP 16.

IF someone in your group solved nine mazes correctly, you would get GBP 0.

Note: Ties will be broken randomly. Thus, IF two people in your group solve eight mazes correctly we flip a coin to see who gets the GBP 16.

Are there any questions?

Slide 7:

You will get 5 min to solve up to 15 mazes.

Please solve as many mazes as you can

Please make sure you have put your name and seat on the maze packet now.
Do not begin until I say go!
OK → GO!
OK → STOP
No Talking!

Slide 8:

In this round you choose between two options.

Option 1: Get GBP 0.50 per maze you solve correctly.

Option 2: Get GBP 2 per maze you solve correctly IF you solve more mazes correctly than the other three people in your group did LAST round.

Example: Say you solve eight mazes correctly this round.

If you chose option 1 you get GBP 4.

If you chose option 2:

You get GBP 16 IF the other three people in your group solved fewer than eight mazes correctly in Round 2.

You get GBP 0 IF one other person solved nine mazes correctly in Round 2.

Note: Ties will be broken randomly. Thus IF one person in your group solved eight mazes correctly in Round 2 we flip a coin to see if you get the GBP 16.
Are there any questions?

Slide 9:²⁸

A supervisor will now come by and give you a card for you to circle Option 1 or Option 2.

Option 1: Get GBP 0.50 per maze you solve correctly.

Option 2: Get GBP 2 per maze you solve correctly IF you solve more mazes correctly than the other three people in your group did LAST round.

Circle your choice, fold the paper and give it back to the supervisor.

You need to write your seat number on the piece of paper

Do not tell anyone your choice!

You will get 5 min to solve up to 15 mazes.

Please solve as many mazes as you can

Do not begin until I say go!

Please make sure you have put your name and seat on the maze packet now.

Do not begin until I say go!

OK → GO!

OK → STOP

No Talking!

Slide 10:

In this round you will not have to do any mazes.

Everyone will be given £5 to play with. Think of the £5 as already being your own money.

You now face a choice:

Option One: Keep your £5.

Option Two: Gamble with your £5.

IF you choose option two you will flip a coin at the end of this round.

IF the coin comes up heads you will get £11.

IF the coin comes up tails you will get £2.

²⁸ We have not yet analysed the results of this fourth round.

A supervisor will now hand you a piece of paper. Choose Option One or Option Two and then fold the paper.
 Please put your seat number on the option card
 Do not tell anyone your choice!
 Everyone will now stand up when the supervisor comes to you and Flip a coin. Your supervisor will record the flip.

Slide 11:

Thank you for completing the mazes!
 Your last set of mazes will now be collected – please stay seated.
 I will now pull the number from the hat..... AND!?
 You will be handed a survey – Read the questions very carefully and make sure you respond to ALL the questions including the ones at the very end.
 After everyone is done completing the survey a supervisor will hand you some refreshments.
 Make sure you put your seat on the survey!
 Then after 10–15 min, your supervisor will give you an envelope with your money and ask you to sign a piece of paper. Then you will go to your bus.
 Please keep your winnings confidential.
 THANKS!
Comment:
 Due to the time it took to fill all the envelopes with money, subjects ended up receiving the money two days later as the students needed to get back to their schools to be picked up by their parents.

Table A1
Dependent Variable (=1) If Student Attends Single-Sex School

Variable	[1]
Essex (=1)	0.84*** (0.05)
Essex × less than average travel time to single-sex school (min)	−0.03*** (0.01)
Essex × more than average travel time to single-sex school (min)	−0.02*** (0.00)
Female (=1)	0.00 (0.07)
All-girls group (=1)	0.00 (0.06)
All-boys group (=1)	0.09 (0.07)
Constant	0.02 (0.04)
Observations	260
R ²	0.476

Note. **p < 0.01, ***p < 0.05, *p < 0.1. Robust standard errors in parenthesis.

ANU and University of Essex
University of Essex

References

- Akerlof, G. and Kranton, R. (2000). 'Economics and identity', *Quarterly Journal of Economics*, vol. 115, pp. 715–53.
- American Association of University Women. (1992). *How Schools Shortchange Girls*, Washington, DC: American Association of University Women.
- Boheim, R. and Taylor, M. (2002). 'Tied down or room to move? Investigating the relationship between housing tenure, employment status and residential mobility in Britain', *Scottish Journal of Political Economy* vol. 49, pp. 369–92.
- Booth, A. and Katic, P. (2011). 'Cognitive skills and risk preferences', mimeo, Australian National University.
- Booth, A. and Nolen, P. (2009). 'Gender differences In competition: the role of single-sex education', CEPR Discussion Paper No. 7214, London, forthcoming *Journal of Economic Behavior and Organisation*.
- Brutsaert, H. (1999). 'Coeducation and gender identity formation: a comparative analysis of schools in Belgium', *British Journal of Sociology of Education*, vol. 20, pp. 343–53.
- Burks, S., Carpenter, J., Goette, L. and Rustichini, A. (2009). 'Cognitive skills affect economic preferences, strategic behavior, and job attachment', *Proceedings of the National Academy of Sciences of the USA*, vol. 106, pp. 7745–50.
- Coleman, J. (1961). *The Adolescent Society*, New York: Free Press.
- Datta Gupta, N., Poulsen, A. and Villeval, M.-C. (2005). 'Male and female competitive behavior: experimental evidence', Working Paper No. 1833, IZA.
- Dohmen, T., Falk, A., Huffman, D. and Sunde, U. (2006). 'The intergenerational transmission of risk and trust attitudes', IZA Discussion Paper No. 2380, forthcoming *Review of Economic Studies*.
- Dohmen, T., Falk, A., Huffman, D., Schupp, J., Sunde, U. and Wagner, G. (2011). 'Individual risk attitudes: new evidence from a large, representative, experimentally-validated survey', *Journal of the European Economic Association*, vol. 9, pp. 522–50.
- Dohmen, T., Falk, A., Huffman, D. and Sunde, U. (2010). 'Are risk aversion and impatience related to cognitive ability?', *American Economic Review*, vol. 100, pp. 1238–60.
- Eckel, C. and Grossman, P. (2002). *Differences in the Economic Decisions of Men and Women: Experimental Evidence*, Amsterdam: Elsevier Science B.V. North-Holland.
- Gneezy, U., Leonard, K. and List, J. (2009). 'Gender differences in competition: evidence from a matrilineal and a patriarchal society', *Econometrica*, vol. 77, pp. 1637–64.
- Gneezy, U., Niederle, M. and Rustichini, A. (2003). 'Performance in competitive environments: gender differences', *Quarterly Journal of Economics*, vol. 118, pp. 1049–74.
- Gneezy, U. and Rustichini, A. (2004). 'Gender and competition at a young age', *American Economic Review Papers and Proceedings*, vol. 94, pp. 377–81.
- Harbaugh, W., Krause, K. and Vesterlund, L. (2002). 'Risk attitudes of children and adults: choices over small and large probability gains and losses', *Experimental Economics*, vol. 5, pp. 53–84.
- Holt, C. and Laury, S. (2002). 'Risk aversion and incentive effects', *American Economic Review*, vol. 92, pp. 1644–55.
- Kessler, S., Ashenden, D., Connell, R. and Dowsett, G. (1985). 'Gender relations in secondary schooling', *Sociology of Education*, vol. 58, pp. 34–48.
- Lavy, V. and Schlosser, A. (2011). 'Mechanisms and impacts of gender peer effects at school', *American Economic Journal: Applied Economics*, vol. 3, pp. 1–33.
- Lawrence, P. (2006). 'Men, women, and ghosts in science', *PLoS Biology*, vol. 4, pp. 13–15.
- Maccoby, E. (1990). 'Gender and relationships: a developmental account', *American Psychologist*, 45, pp. 513–20.
- Maccoby, E. (1998). *The Two Sexes: Growing up Apart, Coming Together*, Cambridge, MA: Harvard University Press.
- Niederle, M. and Vesterlund, L. (2007). 'Do women shy away from competition? Do men compete too much?', *Quarterly Journal of Economics*, vol. 122, pp. 1067–101.
- Niederle, M. and Yestrumskas, A. (2008). 'Gender differences in seeking challenges: the role of institutions', Working Paper No. 13922, NBER.
- Rabin, M. and Weizsäcker, G. (2009). 'Narrow bracketing and dominated choices', *American Economic Review*, vol. 99, pp. 1508–43.
- Sadker, M., Sadker, D. and Klein, S. (1991). 'The issue of gender in elementary and secondary education', *Review of Research in Education*, vol. 17, pp. 269–334.
- Schubert, R., Brown, M., Gysler, M. and Brachinger, H. (1999). 'Financial decision-making: are women really more risk-averse?', *American Economic Review*, vol. 89, pp. 381–5.
- Steele, C., Spencer, S. and Aronson, J. (2002). 'Contending with group image: the psychology of stereotype and social identity threat', *Advances in Experimental Social Psychology*, vol. 34, pp. 379–440.